

PRST (01/03/2021)

$$P(A \cup B) \leq P(A) + P(B).$$

$$\begin{aligned} P\left(\bigcup_{i=1}^k A_i\right) &= P\left(\underbrace{\bigcup_{i=1}^{k-1} A_i}_{\text{}} \cup \underbrace{A_k}_{\text{}}\right) \\ &\leq \underbrace{P\left(\bigcup_{i=1}^{k-1} A_i\right)}_{\text{}} + P(A_k) \\ &\vdots \\ &\leq \sum_{i=1}^k P(A_i) \end{aligned}$$

Conditional probability :

$$\Omega = \left\{ \overset{\checkmark}{1} \overset{\checkmark}{2} \overset{\checkmark}{3} \overset{\checkmark}{4} \overset{\checkmark}{5} \overset{\checkmark}{6} \overset{\checkmark}{7} \overset{\checkmark}{8} \overset{\checkmark}{9} \overset{\checkmark}{0} \right\} \quad A = \{2\}$$

$$P(A) = \frac{1}{10}$$

$$B = \{2, 3, 4, 5, 6, 8, 9\} \quad P(A|B) = \frac{1}{7}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad P(B|B) = 1$$

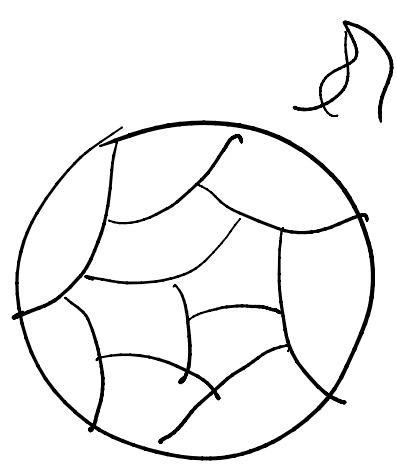
A and B are independent events if

$$P(A|B) = P(A).$$

$$\Rightarrow P(A \cap B) = P(A)P(B).$$

If the sequence of events $\{B_i\}$ in Ω such that they are mutually exclusive (disjoint) and $\bigcup_i B_i = \Omega$.

Then $\{B_i\}$ forms a partition of the sample space Ω .



Result.

$A \subseteq \Omega$, $\{B_i\}$ - partition of Ω .

$$P(A) = \sum_i P(B_i)P(A|B_i)$$

proof

$$\begin{aligned} P(A) &= P(A \cap \Omega) \\ &= P\left(A \cap \bigcup_i B_i\right) \\ &= P\left(\bigcup_i \underbrace{A \cap B_i}_{\text{}}\right) \\ &= \sum_i P(A \cap B_i) \\ &= \sum_i P(B_i)P(A|B_i). \end{aligned}$$

A - system shutdown

B_1 - power disconnect

B_2 - hardware fail

B_3 - software fail

$$\begin{aligned} P(A) &= P(B_1)P(A|B_1) \\ &\quad + P(B_2)P(A|B_2) \\ &\quad + P(B_3)P(A|B_3). \end{aligned}$$

$$= \sum_i P(B_i) P(A|B_i)$$

Bayes theorem

$\{B_i\}$ forms a partition of Ω .
 $A \subseteq \Omega$
 event

$$P(B_i|A) = \frac{P(B_i) P(A|B_i)}{\sum_j P(B_j) P(A|B_j)}$$

proof: $P(B_i|A) = \frac{P(B_i \cap A)}{P(A)} = \frac{P(B_i) P(A|B_i)}{\sum_j P(B_j) P(A|B_j)}$

Ex. MCQ — 3 options (only 1 option is correct).
 If a student knows the answer then the probability of giving right answer is 1.

If he does not know, the probability of giving correct answer is $\frac{1}{3}$.

$\Rightarrow$ The probability that a student knows the correct answer P .

$P(\text{student knows the answer} | \text{given correct answer})$.

$$P(K|C) = \frac{P(K) P(C|K)}{P(K) P(C|K) + P(NK) P(C|NK)}$$

$$= \frac{P}{P + (1-P) \times \frac{1}{3}}$$

Result: If A and B are independent events then

(i) $P(A \cap B^c) = P(A) P(B^c)$
 (ii) $P(A^c \cap B^c) = P(A^c) P(B^c)$

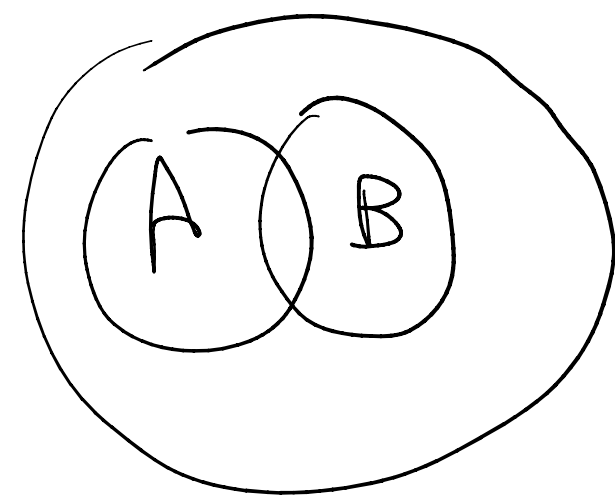
$P(A \cap B) = P(A) P(B)$

proof (i).

$$P(A \cap B^c) = P(A) - P(A \cap B)$$

$$= P(A) - P(A) P(B)$$

$$= P(A) P(B^c)$$



$P(A \cap B) = P(AB)$

$$\begin{aligned}
 \textcircled{ii} \quad P(A^c \cap B^c) &= P((A \cup B)^c) \\
 &= 1 - P(A \cup B) \\
 &= 1 - P(A) - P(B) + P(A \cap B) \\
 &= \frac{1 - P(A) - P(B) + P(A)P(B)}{1} \\
 &= P(A^c) - P(B)P(A^c) \\
 &= P(A^c)P(B^c).
 \end{aligned}$$

Result : $P(A \cap B | C) = P(B | C) P(A | B \cap C).$

Mutually independent
 $A_1, A_2, \dots, A_k$

$$\begin{aligned}
 P(A_{i_1} \cap A_{i_2}) \\
 &= P(A_{i_1})P(A_{i_2})
 \end{aligned}$$

$$\begin{aligned}
 P(A_{i_1} \cap A_{i_2} \cap A_{i_3}) \\
 &= P(A_{i_1})P(A_{i_2})P(A_{i_3})
 \end{aligned}$$

$$\begin{aligned}
 &\vdots \\
 P(A_1 \cap A_2 \cap \dots \cap A_k) \\
 &= P(A_1)P(A_2) \dots P(A_k).
 \end{aligned}$$

Pairwise independent
 $A_1, A_2, \dots, A_k$

$$P(A_{i_1} \cap A_{i_2}) = P(A_{i_1})P(A_{i_2}).$$

$$P(A_1 \cap A_2) = P(A_1)P(A_2)$$

$$(P(A_1 \cap A_2))$$

Ex $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}$
 $P(\omega_1) = \frac{1}{4}$ $P(\omega_2) = \frac{1}{4}$ $P(\omega_3) = \frac{1}{4}$ $P(\omega_4) = \frac{1}{4}$
 $A_1 = \{\omega_1, \omega_2\}$ $A_2 = \{\omega_1, \omega_3\}$
 $A_3 = \{\omega_1, \omega_4\}$ $A_1 \cap A_2 = \{\omega_1\}$

$$P(A_i) = \frac{1}{2} \quad i = 1, 2, 3$$

$$P(A_1 \cap A_2) = \frac{1}{4} = P(A_1)P(A_2)$$

$$P(A_1 \cap A_3) = \frac{1}{4} = P(A_1)P(A_3)$$

$$P(A_2 \cap A_3) = \frac{1}{4} = P(A_2)P(A_3)$$

pairwise independent

$$P(A_1 \cap A_2 \cap A_3) = \frac{1}{4} \quad \text{not mutually independent.}$$

$$\neq P(A_1)P(A_2)P(A_3) = \frac{1}{8}$$

Ex. What is the probability that all 3 children in a family have different birthdays.

Sol - $P = \frac{365 \times 364 \times 363}{365 \times 365 \times 365}$

Ex2. A batch of products contains 10 products of which 4 are defective. If 3 products are chosen randomly, L is the probability that

What is the
 none of them will be defective?

Sol.

$$\frac{{}^6C_3}{{}^{10}C_3} = \frac{1}{6}$$

Ex Two players A, B throw a die alternatively.
 The who first gets a six will win.
 If A begins, then what is the prob that
 A will win?

E - getting 6
 F - not 6

$$P(\text{A will win}) = P(E) + P(FFE) + P(FFFE) + \dots$$

$$= \frac{1}{6} + \left(\frac{5}{6}\right)^2 \frac{1}{6} + \left(\frac{5}{6}\right)^4 \frac{1}{6} + \dots$$

$$= \frac{1}{6} \left(1 + \left(\frac{5}{6}\right)^2 + \left(\frac{5}{6}\right)^4 + \dots \right)$$

$$= \frac{\frac{1}{6}}{1 - \left(\frac{5}{6}\right)^2} = \frac{6}{11}$$